

Five-variable k-map-

A five-variable (A, B, C, D, E) can have $2^5 = 32$ possible combinations of input combinations

such as:- $\bar{A}\bar{B}\bar{C}\bar{D}\bar{E}, \bar{A}\bar{B}\bar{C}\bar{D}E, \dots, ABCDE$ $m_0, m_1, \dots, m_{31}$
 $A+B+C+D+E, A+B+C+D+\bar{E}, \dots, \bar{A}+\bar{B}+\bar{C}+\bar{D}+\bar{E}$ $M_0, M_1, \dots, M_{31}$

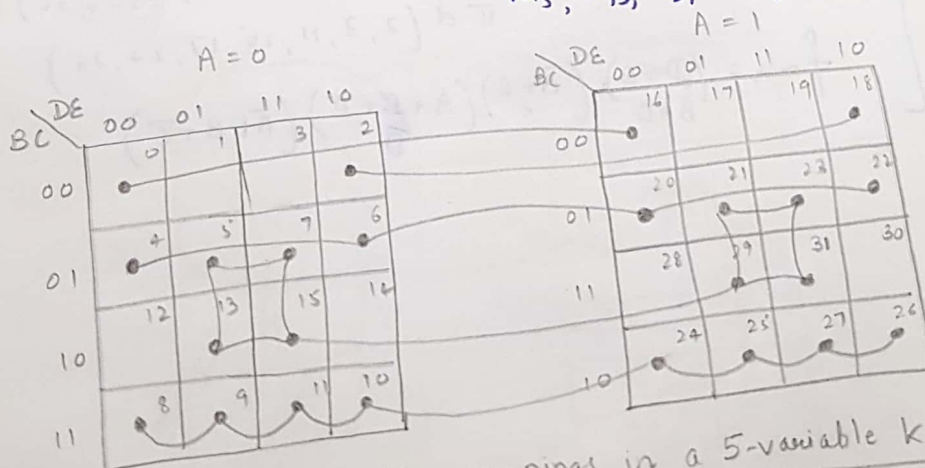
* The 32 squares of the k-map are divided into 2 blocks of 16 squares each.

* The left block represents minterms from m_0 to m_{15} in which $A=0$ & the right block represents minterms from m_{16} to m_{31} in which $A=1$.

* The five variable k-map may contain 2-squares, 4-squares, 8-squares, 16-squares or 32-squares involving these 2 blocks. Squares are also considered adjacent in these two blocks, if when superimposing one block on top of another, the squares coincide with one another.

Some possible 2-squares are:- $m_0, m_{16}; m_2, m_{18}; m_5, m_{21}; m_{15}, m_{31}, m_{11}, m_{27}$.

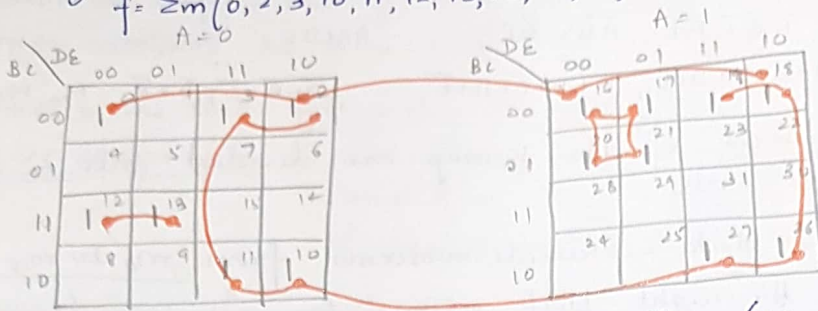
Some possible 4-squares are:- m_0, m_2, m_{16}, m_{18}
 m_0, m_1, m_{16}, m_{17}
 m_0, m_4, m_{16}, m_{20}
 $m_{13}, m_{15}, m_{29}, m_{31}$
 $m_5, m_{13}, m_{21}, m_{29}$



Some possible groupings in a 5-variable k-map

Ques. Reduce the following expression in SOP & POS forms using mapping:-

$$f = \sum m(0, 2, 3, 10, 11, 12, 13, 16, 17, 18, 19, 20, 21, 26, 27)$$



$$f_{\min} = \bar{A}BC\bar{D} + \bar{B}\bar{C}E + A\bar{B}\bar{D} + \bar{C}D \quad (16 \text{ inputs})$$

In POS form,

$$f = \prod M(1, 4, 5, 6, 7, 8, 9, 14, 15, 22, 23, 24, 25, 28, 29, 30, 31)$$

$$f_{\min} = (A+B+D+E)(A+B+\bar{C})(\bar{B}+C+D)(\bar{A}+\bar{B}+D)(\bar{C}+\bar{D}) \quad (20 \text{ inputs})$$

Ques. Reduce the following expression to the simplest possible POS and SOP forms.

$$\left\{ \begin{aligned} f &= \sum m(6, 9, 13, 18, 19, 25, 27, 29, 31) + d(2, 3, 11, 15, 17, 24, 28) \\ f_{\min} &= BE + \bar{B}\bar{C}D + A\bar{B}D\bar{E} \end{aligned} \right.$$

$$\left\{ \begin{aligned} f &= \prod M(0, 1, 4, 5, 7, 8, 10, 12, 14, 16, 20, 21, 22, 23, 26, 30) \\ &\quad \prod d(2, 3, 11, 15, 17, 24, 28) \\ f_{\min} &= (\bar{D}+\bar{E})(\bar{B}+\bar{E})(A+\bar{B}+\bar{E})(\bar{A}+B+\bar{C}) \end{aligned} \right.$$

Six-variable k-map -

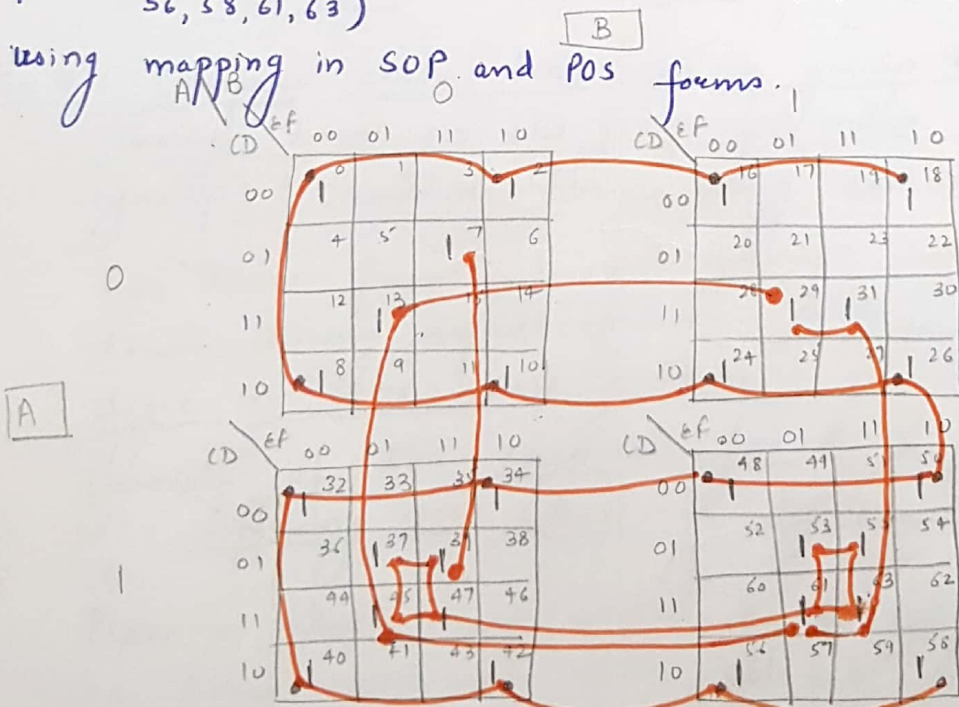
A 6 variable (A, B, C, D, E, F) expression can have $2^6 = 64$ possible combinations of input variables such as $\bar{A}\bar{B}\bar{C}\bar{D}\bar{E}\bar{F}$, $\bar{A}\bar{B}\bar{C}\bar{D}\bar{E}F$, ..., $ABCDEF$
 $(A+B+C+D+E+F)$, $(A+B+C+D+E+\bar{F})$, ..., $(\bar{A}+\bar{B}+\bar{C}+\bar{D}+\bar{E}+\bar{F})$

- * The 64-squares of k-map are divided into four blocks of 16 squares each. Each square represent a minterm or a maxterm.
- * Values of A & B remain constant for each block.
- * Top left block represents m_0 to m_{15} $A=0, B=0$
 Top right " " " m_{16} to m_{31} $A=0, B=1$
 Bottom left block " " m_{32} to m_{47} $A=1, B=0$
 Bottom right " " m_{48} to m_{63} $A=1, B=1$
- * It may contain 2-squares, 4-squares, 8-squares, 16-squares, 32-squares or a 64-square.
- * Squares are considered adjacent in 2 blocks, if upon superimposing one block on top of another block, that is, above or below or beside the first block, the squares coincide with one another. Diagonal elements like m_{10}, m_{58} ; m_{15}, m_{63} ; m_{18}, m_{34} ; m_{29}, m_{45} are not adjacent to each other.

Ques. Reduce the expression

$$f = \sum m(0, 2, 7, 8, 10, 13, 16, 18, 24, 26, 29, 31, 32, 34, 37, 39, 40, 42, 45, 47, 48, 50, 53, 55, 56, 58, 61, 63)$$

using mapping in SOP and POS forms.



$$f = \bar{B}\bar{C}DEF + CDEF + BCDF + ADF + \bar{D}\bar{F} \quad (23 \text{ inputs})$$

In POS form,

$$f = \prod M(1, 3, 4, 5, 6, 9, 11, 12, 14, 15, 17, 19, 20, 21, 22, 23, 25, 27, 28, 30, 33, 35, 36, 38, 41, 43, 44, 46, 47, 51, 52, 54, 57, 59, 60, 62)$$

$$f_{min} = (A+B+\bar{C}+\bar{D}+\bar{E})(\bar{D}+F)(\bar{D}+\bar{F})(A+\bar{B}+C+\bar{F})(A+C+\bar{D}+E)$$

(22 inputs)

Limitations of Karnaugh Maps -

1. Good for expressions with less than 6 variables. Difficult in case of more variables.
2. K-map simplification is a manual technique; heavily dependent on human abilities. It cannot be programmed.

$\therefore$, an exact tabular method to simplify the Boolean expressions was developed called the

Quine-McCluskey Method / Tabular Method.